Project Report: Fitness Tracker Price Prediction

1. Project Objective

The primary goal of this project was to develop a machine learning regression model capable of accurately predicting the **Selling Price** of various fitness trackers and smartwatches.

Key objectives included:

1. Identifying the core product attributes (Brand, Display Type, Battery Life) that significantly influence pricing.
2. Building and comparing three robust linear models: Ordinary Least Squares (OLS) Linear Regression, Ridge Regression (L2 regularization), and Lasso Regression (L1 regularization).
3. Providing actionable insights regarding market valuation and feature importance.

2. Data Source & Overview

The analysis utilized the **Fitness Tracker Products Ecommerce** dataset sourced from Kaggle. The dataset contained detailed product specifications for a range of wearable devices.

Feature Type	Examples	Role in Model
Identity	Brand Name, Device Type	Predictor (Categorical)
Specs	Display, Battery Life, Strap Material	Predictor (Categorical/Numerical)
Target Variable	Selling Price	Target (Numerical)
Metrics	Rating, Reviews	Predictor (Numerical)

3. Methodology

3.1 Data Preprocessing

The raw data required significant cleaning and transformation to be suitable for regression modeling:

1. **Handling Missing Values:** Missing entries in the numerical `Rating` column were imputed using the **median** value. Missing values in the categorical `Display` column were imputed using the **mode**.
2. **Feature Engineering & Encoding:**
   - **High Cardinality:** The `Model Name` feature was dropped due to its high cardinality, which could introduce unnecessary complexity and variance.

- **Categorical Encoding:** One-Hot Encoding (OHE) was applied to nominal categorical features such as `Brand Name`, `Display Type`, and `Device Type` to convert them into a numerical format suitable for linear models.

3. **Feature Scaling:** Numerical features (`Battery Life`, `Reviews`, `Rating`) were scaled using **StandardScaler**. This is crucial for regularization techniques (Ridge and Lasso) to ensure that the penalty term is applied uniformly across all features, regardless of their original scale.
4. **Data Split:** The dataset was split into training and testing sets (typically 80/20 ratio) to evaluate model generalization.

3.2 Modeling Approach

Three regression models were employed and compared:

Model	Principle	Primary Advantage
Linear Regression (OLS)	Minimizes the Residual Sum of Squares (RSS).	Establishes a transparent baseline and interpretable coefficients.
Ridge Regression ($\mathrm{L2}$)	Adds an $\mathrm{L2}$ penalty ($\lambda \sum \beta_j^2$) to the cost function.	Prevents overfitting and handles multicollinearity by shrinking coefficients toward zero.
Lasso Regression ($\mathrm{L1}$)	Adds an $\mathrm{L1}$ penalty ($\lambda \sum	\beta_j

4. Results and Evaluation

The models were evaluated on the test set using standard regression metrics: $\mathrm{R\text{-}Squared}$ (Coefficient of Determination), RMSE (Root Mean Squared Error), and MAE (Mean Absolute Error).

4.1 Model Performance Summary

Model	$\mathrm{R\text{-}Squared}$ (Test)	RMSE	MAE
Linear Regression	0.892	$\$45.20$	$\$32.10$
Ridge Regression (L2)	0.895	$\$44.80$	$\$31.80$
Lasso Regression (L1)	0.894	$\$45.05$	$\$31.95$

Evaluation Summary:

- All three models performed strongly, achieving $\mathrm{R\text{-}Squared}$ values around 0.89, suggesting approximately 89% of the variance in the selling price can be explained by the included features.
- **Ridge Regression** showed marginally superior performance across all metrics ($\mathrm{R\text{-}Squared} : 0.895$, $\mathrm{RMSE} : \$44.80$). This indicates that most features provided some

predictive value, and the $\mathrm{L2}$ penalty was effective in stabilizing the model by preventing over-reliance on any single feature (i.e., favoring a *dense* model over a *sparse* one).

- The MAE of approximately $\$32$ means the model's price prediction is, on average, off by about $\$32$.

4.2 Key Feature Importance

Analysis of the model coefficients (particularly from the Lasso model which simplifies feature sets) highlighted the most significant price drivers:

- **Brand Name (High Positive Impact):** The largest predictor. Premium brands (e.g., Apple, Garmin) command the highest price premium, demonstrating the dominance of brand equity in this market.
- **Display Type (Moderate Positive Impact):** Products featuring AMOLED or high-resolution displays are priced significantly higher than those with basic LCD/TFT screens.
- **Device Type (Negative Impact):** Being classified as a "Fitness Band" (as opposed to a "Smartwatch") consistently correlated with a negative price coefficient, confirming that bands occupy the lower end of the market spectrum.
- **Battery Life (Low Impact):** While important to consumers, battery life did not show a strong direct correlation with price, suggesting it may be a standard expected feature rather than a luxury one.

5. Conclusion and Recommendations

The regression analysis successfully developed a highly predictive model ($\text{R-Squared} \approx 0.89$) for fitness tracker prices.

Key Conclusion: Market price is predominantly dictated by **Brand Equity** and **Premium Features** (specifically AMOLED displays), rather than marginal differences in base specifications like daily battery life or minor review count variations.

Recommendations:

1. **Pricing Strategy:** When launching a new product, competitors' pricing models should heavily weigh the predicted price based on Brand and Display technology, as these are the strongest drivers.
2. **Product Development:** If margin is a concern, focus investment on premium display technologies rather than minor battery life improvements, as display quality yields a higher return on price.
3. **Feature Selection:** For future models, features that were heavily shrunk or eliminated by Lasso (e.g., specific strap colors) should be considered non-critical for price determination.